

Analysis of a Tripartite Qubit-Magnon-Cavity system using COMSOL

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Hybrid quantum systems combine superconducting circuits with other quantum excitations to exploit the strengths of each subsystem. Crucially, superconducting qubits are leading the race to establish modern quantum computers because of their scalability and ability to couple to other systems for storing and reading out states whenever required. In this report, I will be analyzing a qubit-magnon-cavity system with the help of COMSOL Multiphysics. This report is divided into 3 sections and is inspired by Tabuchi's paper[1]. Section I contains qubit-cavity coupling. Section II contains magnon-cavity coupling and Section III contains the tripartite system. Let us now examine each of them in detail.

1 Cavity-Qubit System

A microwave cavity acts like a structure that confines field within a small region of space and is generally used in cQED systems. These cavities are widely used in quantum information systems because they can store microwave photons for a finite amount of time and allow controlled interactions with other quantum devices. When a superconducting qubit is placed inside or coupled to such a cavity, the two systems can exchange energy through microwave photons. The frequency of such a rectangular cavity with dimensions a,b and d is given by the following equation,

$$f_{mnl} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{l}{d}\right)^2} \quad (1)$$

where f_{mnl} is given as the cavity frequency corresponding to the mode indices. For example, f_{101} is the fundamental cavity frequency and is used in this report. The dimensions of the cavity are $a = 40$, $b = 10$, $d = 32$ (all mm), such that the theoretical frequency is 6.002 GHz. The electric and magnetic fields for this mode is given in Figure 1.

The electric field and magnetic spots give us an idea about the locations of maximum coupling between the electric field and the qubit as well as the magnetic field and the magnon. The qubits and YIG film used as the magnon were placed accordingly. The bare cavity frequency was 5.9988 GHz found out from the COMSOL eigenmode solver.

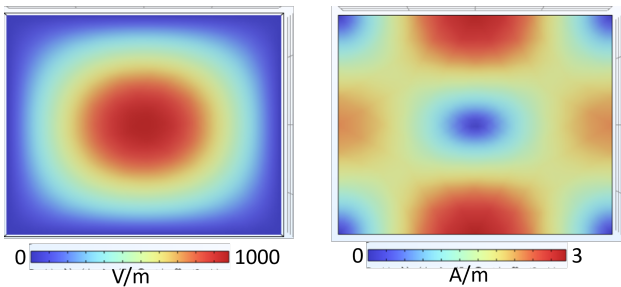


Figure 1: The electric field plot(left) and the magnetic field intensity plot(right) of the fundamental mode.

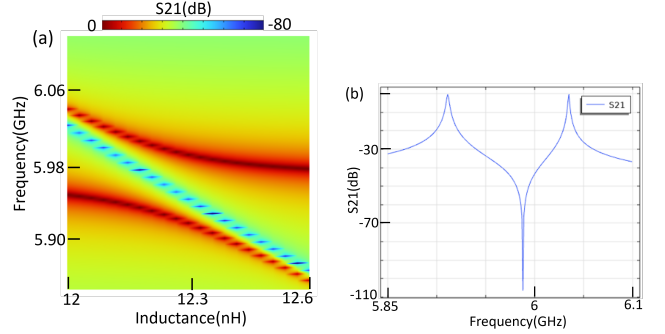


Figure 2: (a) The S_{21} plot obtained by varying inductance. The blue diagonal line arises due to antiresonance. (b) The S_{21} graph curve for a specific inductance. The dip corresponds to the antiresonance, while the peaks correspond to the resonant transmission features.

The superconducting qubit was placed at the centre of the cavity. The dimensions of the capacitor pads are $0.3\text{mm} \times 0.3\text{mm}$ while the inductor dimensions were $20\mu\text{m} \times 60\mu\text{m}$. It was positioned on top of a silicon substrate and a sapphire place holder. The capacitor pads were assigned as perfect electric conductors, and a 12.3nH lumped element was assigned to the connection pad which acted as the inductor. For measuring frequencies a cylindrical probe(Fig. 2(a)) was used with input pin depth of 0.8mm . The external coupling and corresponding quality factors depend on the probe pin depth. Initially an eigenmode simulation was done to extract the resonant frequencies and the Quality Factors were evaluated with the help of that. The values are given in Table 1.

Table 1: System parameters

Frequency (GHz)	Quality Factor (Q)	κ (MHz)
5.8592	16310.89	0.359
5.9620	1779.2	3.35

The silicon and sapphire were modelled without material losses, so the corresponding internal loss terms were set to zero in the simulation. A curve fitting was done for the S_{21} plot and the coupling strength extracted turned out to be 40MHz . The cooperativity turned out to be 1364.6 which places the system in strong coupling regime. When the qubit frequency is far detuned from the central cavity mode, it avoids resonant hybridization but undergoes radiative relaxation mediated by the leaky cavity ports (the Purcell effect).

Given a large detuning $\Delta = 1\text{GHz} = 1000\text{MHz}$ and taking the dominant total cavity decay rate $\kappa = 3.35\text{MHz}$, the induced Purcell relaxation rate Γ_P is defined as:

$$\Gamma_P = \kappa \left(\frac{g}{\Delta}\right)^2 \quad (2)$$

Substituting the system values:

$$\Gamma_P = 3.35 \text{ MHz} \times \left(\frac{40 \text{ MHz}}{1000 \text{ MHz}} \right)^2 \quad (3)$$

$$\Gamma_P = 3.35 \times (0.04)^2 = 3.35 \times 0.0016 = 0.00536 \text{ MHz} \quad (4)$$

The corresponding qubit relaxation lifetime T_1 is given by the inverse of the relaxation rate:

$$T_1 = \frac{1}{\Gamma_P} = \frac{1}{0.00536 \times 10^6 \text{ s}^{-1}} \approx 186.57 \text{ } \mu\text{s} \quad (5)$$

This calculation shows that, at large detuning, the qubit is well protected from cavity losses, giving a T_1 of about 186.57 μs .

1.1 Observation of Antiresonance

Anti-resonance is commonly observed in coupled oscillator systems and in engineering terms, it is the point where the reactance vanishes but the resistive impedance is high, approaching infinity[2]. If we consider our system as two harmonic oscillators coupled with coupling strength of g with one oscillator driven by an oscillating external force F , we can write two differential equations (6) to solve for the system.

$$\begin{aligned} \ddot{x}_1 + 2\gamma_1\dot{x}_1 - 2g\omega_1x_2 + \omega_1^2x_1 &= 2F \cos(\omega t), \\ \ddot{x}_2 + 2\gamma_2\dot{x}_2 - 2g\omega_2x_1 + \omega_2^2x_2 &= 0. \end{aligned} \quad (6)$$

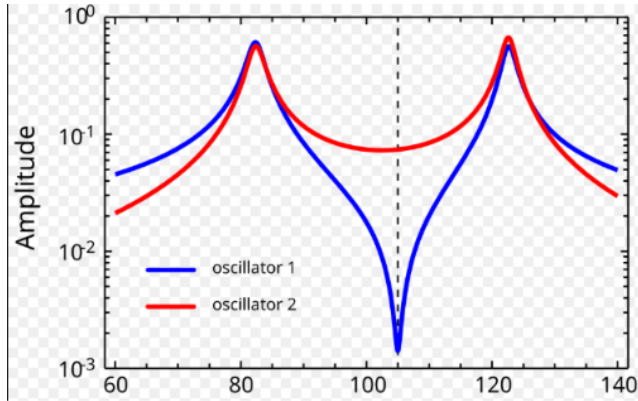


Figure 3: The plot shows amplitude vs frequency for a coupled system.

The antiresonance in our system arises because the direct microwave transmission path and the coherent scattering path from the qubit destructively interfere, strongly suppressing the transmitted signal. Unlike the hybridized upper and lower polariton branches, which bend and split due to strong coupling, this narrow dip closely tracks the bare qubit frequency and can provide a relatively direct indication of the qubit frequency. This makes it a reliable way to read off the qubit frequency directly from the S_{21} data.

In practice this effect is hard to use because it's very sensitive to fabrication and can't be tuned easily, so it hasn't seen much use in real quantum processors. Even small variations can weaken the effect. In addition, the anti-resonance frequency is not easily adjustable. If the qubit frequency needs to be tuned for quantum operations, the anti-resonance may no longer align with it, making the overall system more difficult to design and operate.

2 Cavity-Magnon System

A magnon is the collective excitation of spin particles in a magnetic material. They have the ability to interact with a microwave photon, making them highly useful for storing and transferring quantum information in hybrid quantum systems.

In the presence of an external magnetic field, the rate of change of magnetization is governed by the Landau-Lifshitz-Gilbert equation [3] given as,

$$\frac{d\mathbf{M}}{dt} = -\gamma(\mathbf{M} \times \mathbf{H}_{\text{eff}}) + \frac{\alpha_G}{M_s} \left(\mathbf{M} \times \frac{d\mathbf{M}}{dt} \right) \quad (7)$$

In this equation, $\mathbf{M}$ is the magnetization vector, $\mathbf{H}_{\text{eff}}$ is the effective magnetic field, and γ is the gyromagnetic ratio. The parameter α_G is the dimensionless Gilbert damping constant, which represents the internal magnetic loss of the YIG material. To define the YIG material in COMSOL, the Polder tensor[3] was used, which is given as

$$\boldsymbol{\mu}_r = \begin{pmatrix} \mu & i\kappa & 0 \\ -i\kappa & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (8)$$

Where,

$$\mu = 1 + \frac{\omega_m(\omega_0 + i\alpha_G\omega)}{(\omega_0 + i\alpha_G\omega)^2 - \omega^2} \quad (9)$$

$$\kappa = \frac{\omega_m\omega}{(\omega_0 + i\alpha_G\omega)^2 - \omega^2} \quad (10)$$

where $\omega_0 = \gamma H_0$ is the Larmor precession frequency and $\omega_m = \gamma M_s$ is the characteristic frequency determined by the saturation magnetization M_s of the ferromagnet, α_G is the Gilbert damping constant taken as 0.0001 for simulation.

In our system, a thin film YIG acted as the magnon. The dimensions of the film were 3mm*3mm*0.1mm where the smaller length is along the y axis. The external bias field was applied perpendicular to the direction of cavity's magnetic field for maximum interaction between the cavity and magnon. The magnon was strategically placed at the point of maximum magnetic field in the cavity and the frequency of the magnon followed the Kittel equation[3] given as

$$\omega = \gamma\mu_0\sqrt{H_0(H_0 + M_s)} \quad (11)$$

In this equation, ω is the angular ferromagnetic resonance (FMR) frequency, γ is the gyromagnetic ratio, μ_0 is the vacuum permeability, H_0 is the applied external static magnetic field(A/m), and M_s is the saturation magnetization of the material(A/m).

The magnetic field was varied from 0.136 T to 0.140 T and the mode splitting was observed as shown in Figure 4. The coupling strength g was extracted from curve fitting and was found out to be 35 MHz. The coupling strength g depends on the size of the film because a larger film contains more participating spins and can interact more strongly with the cavity magnetic field. The anti-resonance dip is found in this case as well due to the magnon being completely out of phase (180°) with the field of the cavity causing the destructive interference.

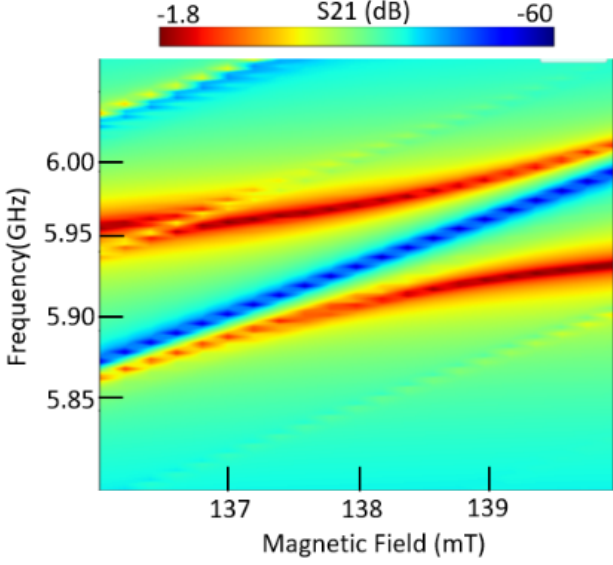


Figure 4: Mode splitting for a cavity-magnon system with 35MHz coupling strength.

3 The Tripartite System

Combining a superconducting qubit and a tunable magnon system in one cavity lets us use the fast processing of qubits along with the long coherence times of magnons. The cavity mediates the interaction between the qubit and magnon, leading to three-way hybridization when all three modes are brought close in frequency. The physical layout of this integrated system is shown in Figure 5

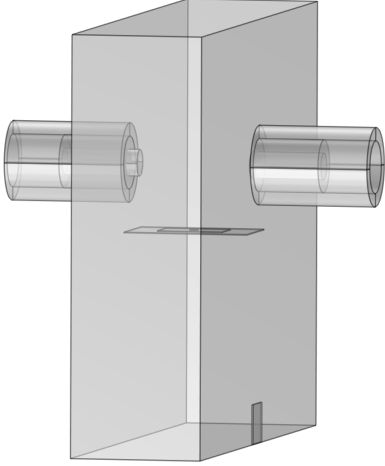


Figure 5: The tripartite system with qubit and magnon integrated inside the vacuum.

As the qubit at the center of the cavity and the YIG film at the bottom are physically separated, their direct mutual coupling is negligible ($g_{mq} \approx 0$). This spatial separation makes the direct qubit-magnon coupling negligible in the present model, allowing the interaction to occur primarily through the cavity. Under the standard rotating-wave approximation (RWA), and keeping our probe power low enough that the transmon stays within its linear, low-excitation regime, we can model the entire system's coupled dynamics using a clean, three-mode linear Hamiltonian,

$$\begin{aligned} \mathcal{H}_{sys}/\hbar = & \Delta_c c^\dagger c + \Delta_m m^\dagger m + \Delta_q q^\dagger q \\ & + g_{cm}(c^\dagger m + cm^\dagger) \\ & + g_{cq}(c^\dagger q + cq^\dagger) \\ & + g_{mq}(m^\dagger q + mq^\dagger) \end{aligned} \quad (12)$$

Here, c , m , and q are the standard bosonic annihilation operators for the cavity photons, the YIG magnons, and the linear qubit excitations, respectively. The detunings from the incoming probe tone frequency (ω) are captured by $\Delta_j = \omega_j - \omega$ (where $j \in \{c, m, q\}$). The fixed hardware parameters g_{cm} and g_{cq} define how strongly the magnon and qubit interact with the cavity field, while g_{mq} is left in the equations as a placeholder to account for any microscopic cross-talk. To bridge the gap between our isolated, closed-system Hamiltonian $\hat{H}_{linear}$ and the macroscopic, dissipative scattering matrices (S_{21}) measured by a Vector Network Analyzer (VNA), we apply Collett-Gardiner input-output theory[4].

The time-evolution of the internal cavity field operator $\hat{a}$ under environmental coupling and port dissipation is governed by the quantum Heisenberg-Langevin equation,

$$\dot{\hat{a}} = -\frac{i}{\hbar}[\hat{a}, \hat{H}_{linear}] - \frac{\kappa_{total}}{2}\hat{a} + \sqrt{\kappa_{ext,1}}\hat{a}_{in}(t)$$

where $\hat{a}_{in}(t)$ represents the continuous microwave input driving field at Port 1. This allows us to map out the complete, non-degenerate transmission spectrum (S_{21}) across the entire frequency range.

$$\begin{aligned} S_{21} = & -\sqrt{\kappa_1 \kappa_2} \left[\left(i\Delta_c + \frac{\kappa_c}{2} + \frac{g_{cq}^2}{i\Delta_q + \frac{\gamma_q}{2}} \right) \right. \\ & \left. + \frac{g_{cm}^2}{i\Delta_m + \frac{\kappa_m}{2}} \right]^{-1} \end{aligned} \quad (13)$$

where κ_c ($\kappa_c = \kappa_1 + \kappa_2 + \kappa_{int}$) is the total cavity decay rate, κ_m is the overall magnon loss rate, and γ_q represents the relaxation of our linear qubit mode. This framework reproduces the multi-peak splitting and transparency dips seen in the COMSOL sweeps.

Following the work of Tabuchi [1], where the cavity acted only as a virtual bus to mediate coupling between the qubit and magnon, our approach keeps the cavity as an active part of the hybrid system. Since a low-power microwave signal is used, the average photon number inside the cavity remains very small ($\langle \hat{a}^\dagger \hat{a} \rangle \ll 1$). Under these conditions, the transmon remains mostly in the $|0\rangle$ and $|1\rangle$ states. Hence, it is treated as a linear harmonic oscillator for the analysis of the steady-state S_{21} response. This complete three-mode framework enables us to study the full S_{21} transmission response, reproducing the three-peak polariton splitting and the sharp antiresonance features.

Figure 6 shows the simulated S_{21} transmission as the external magnetic field (H_0) is varied. The magnon mode shifts linearly with the applied magnetic field. As the magnon frequency approaches the cavity and qubit frequencies, all three modes interact and hybridize. Around $H_0 \approx 0.1386$ T, the spectrum splits into three distinct branches, confirming tripartite strong coupling. The sharp transmission dip at the centre further highlights the strong interaction between the three modes.

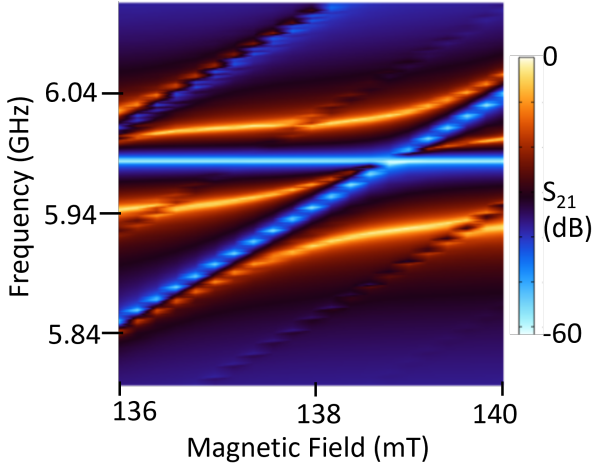


Figure 6: Simulated S_{21} transmission map showing tripartite mode splitting and suppression as the magnetic field H_0 is swept.

To extract the device parameters, the analytical tripartite S_{21} model was fitted to the COMSOL simulation data at a magnetic field of 138.6 mT, as shown in Figure 7. The fitting was performed using the built-in curve fitting tool, and the optimized parameters were extracted from the best fit. The fitted cavity, magnon, and qubit frequencies were found to be $\omega_c/2\pi = 5.9563$ GHz, $\omega_m/2\pi = 5.9768$ GHz, and $\omega_q/2\pi = 5.9582$ GHz, respectively. The fitting also gave coupling strengths of $g_{cm}/2\pi = 30.87$ MHz and $g_{cq}/2\pi = 31.28$ MHz, indicating that the cavity, magnon, and qubit are in the strong coupling regime. The fitted dissipation constants at 138.6 mT were a cavity loss rate of $\kappa_c/2\pi = 3.442$ MHz, a magnon loss rate of $\kappa_m/2\pi = 0.352$ MHz, and a qubit relaxation rate of $\gamma_q/2\pi = 1.000$ MHz. The extracted cavity loss rate closely matches the value of 3.35 MHz reported in Section I, while the qubit and magnon loss rates show small deviations from the previously extracted values.

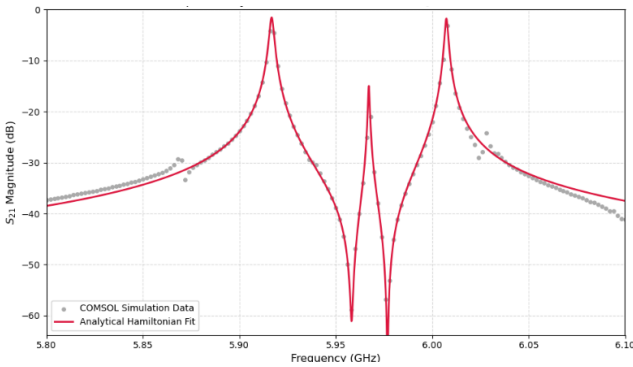


Figure 7: Simulated and analytical S_{21} transmission spectrum at 138.6 mT, highlighting the characteristic hybrid tripartite splitting profile.

The fitted model agrees well with the COMSOL results and reproduces the three hybridized modes along with the transmission dips. A small mismatch can still be seen near the edges of the spectrum, where the analytical curve is slightly different from the simulated response. This is expected because the analytical model considers only three interacting modes, whereas the COMSOL simulation also includes effects such as parasitic capacitances, mesh-related numerical errors, and higher-order Walker modes of the YIG thin film.

The transmission spectrum shows a sharp antiresonance mode, where the S_{21} transmission drops to nearly -60 dB. This is caused by destructive interference between the different excitation pathways in the hybridized tripartite system. When the microwave signal excites both the cavity mode and the localized magnon mode, the two pathways interfere destructively and suppress the transmission through the resonator. Such interference effects can be useful in hybrid quantum-memory schemes, where dark states can reduce the participation of lossy cavity modes.

4 Design Motivation and Insights

The cavity frequency was intentionally chosen close to both the qubit and magnon frequencies so that all three modes could hybridize simultaneously. If the cavity were placed far from resonance, the system would mainly show two independent avoided crossings instead of a true tripartite interaction. The qubit and YIG film were also kept physically separated to make the direct qubit-magnon coupling negligible, allowing the interaction to occur primarily through the cavity. Unlike the work of Tabuchi *et al.*, where the cavity acts mainly as a virtual bus, the cavity in this design remains an active part of the hybrid system and directly contributes to the observed three-mode splitting. These observations suggest that careful choice of the cavity frequency and device geometry is important for realizing controllable hybrid quantum systems for state transfer, quantum memories, and future quantum networks.

5 Summary and Future Outlook

In this work, a linear tripartite cavity-magnon-qubit system was designed and studied in the strong coupling regime. COMSOL simulations together with the Collett-Gardiner input-output theory were used to model the S_{21} response of the system. At the resonant magnetic field of $H_0 = 0.1386$ T, the analytical fitting gave coupling strengths of $g_{cm}/2\pi = 30.87$ MHz and $g_{cq}/2\pi = 31.28$ MHz. The analytical model showed good agreement with the simulation and reproduced the mode splitting and antiresonance.

In future, this design can be extended by using multiple transmons or YIG elements to build larger hybrid quantum systems. Further work should also include the effects of parasitic capacitance, higher-order cavity modes, and magnetic field non-uniformity. Such systems could find applications in quantum memories and quantum routing.

References

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